



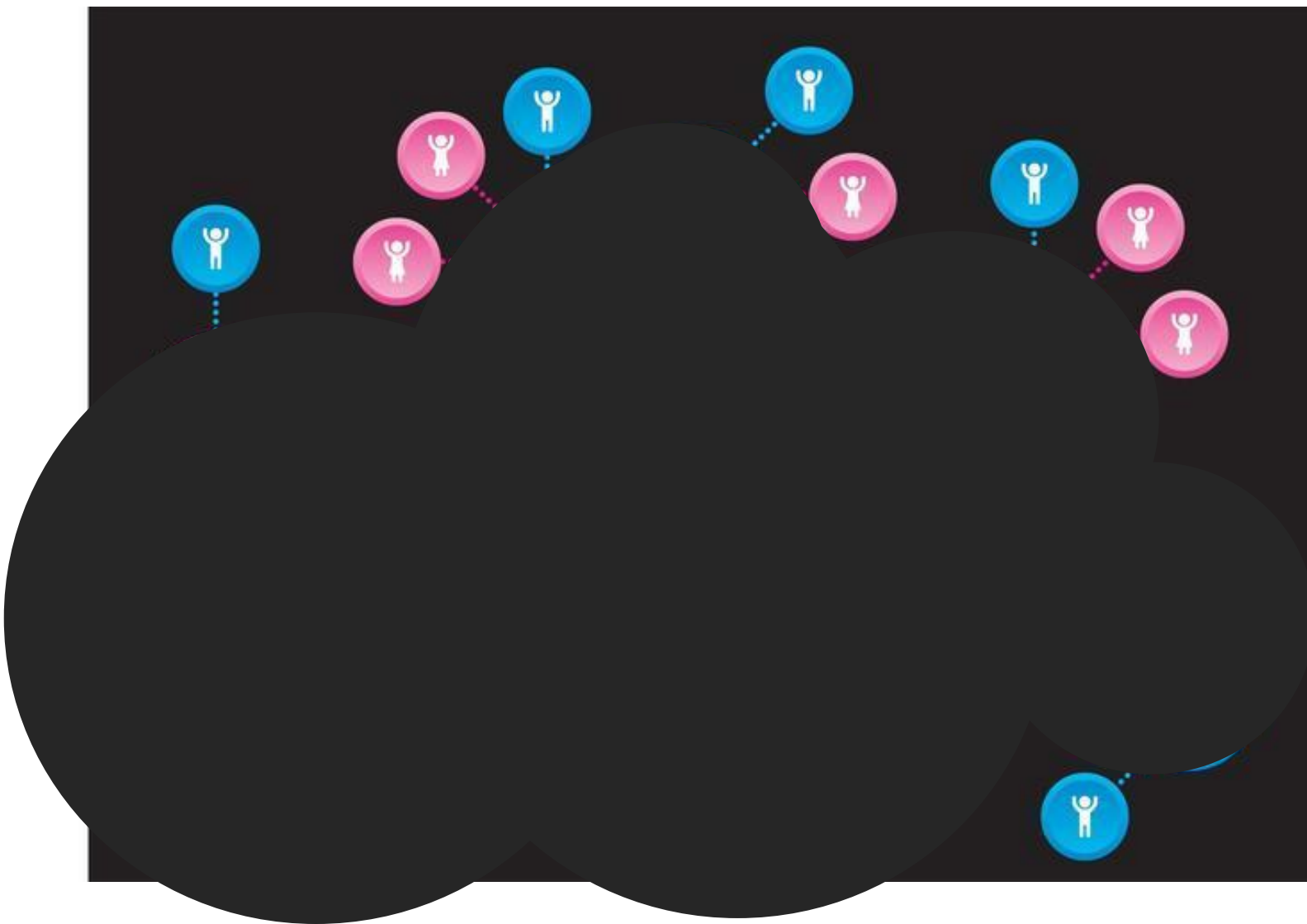
多序列比对的原理以及 clustal在多序列比对中的 应用

内容提要



- 多序列比对的意义
- 多序列比对的方法
- 自动多序列比对的算法
- **Clustalx的使用（clustal法）**
- 实例分析

探索多个生物分子间的关系





多序列比对的作用

- 用于描述一个同源基因之间的亲缘关系的远近，应用到**分子进化分析**中。
- 用于描述一组序列之间的相似性关系，以便了解一个**基因家族**的基本关系特征，寻找**motif**，保守区域等。构建**profile**，打分矩阵等。

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Mus      : MCAAR-----LSAAAQSTVYAFSARPLTGGEFVSLGSLRGKVLLIENVASLUGTTIRDYTEMN
Rattus   : MSAAR-----LSAVAQSTVYAFSARPLAGGEPVSLGSLRGKVLLIENVASLUGTTIRDYTEMN
Cricetulus : MCAAR-----LSAAVQSTVYAFSARPLTGGEFVSLGSLRGKVLLIENVASLUGTTIRDYTEMN
Pan      : MCAAR-----IAAAAQSVYAFSARPLAGGEPVSLGSLRGKVLLIENVASLUGTTVRDYTQMN
Macaca   : MCAAR-----IAAAA---VYAFSARPLAGGEPVSLGSLRGKVLLIENVASLUGTTVRDYTQMN
Equus    : MCAAQ-----IAAAARRSVYAFSARPLAGGEPVSLGSLRGKVLLIENVASLUGTTVRDYTQMN
Bos      : MCAAQRSAAALAAAAPRTVYAFSARPLAGGEPFNLSSLRGKVLLIENVASLUGTTVRDYTQMN
Sus      : MCAAQRSAAALAAVAPRSVYAFSARPLAGGEPISLGLRGKVLLIENVASLUGTTVRDYTQMN
Oryctolagus : MCAAR-----MAAAAQSVYSFSAHPLAGGEPVNLGSLRGKVLLIENVASLUGTTVRDYTQMN
Gallus   : MAATG-----LAGIVARPLGAAEPIALSSLRGKVLLVNVASLUGTTIRDFLQLN

McAa      1 a a      6yafsaRPL ggEP   LgSLRGKVLL6eNVASLUGTT RD5t26N

```



多序列比对的方法

ABCD	A B C D
ABCCD	A B C C D
BCD	B C D

基本上多序列比对可以分为

1.手工比对（辅助编辑软件如**bioedit**, **seaview**, **Genedoc**等）

通过辅助软件的不同颜色显示不同残基，靠分析者的观察来改变比对的状态。

2.计算机程序自动比对

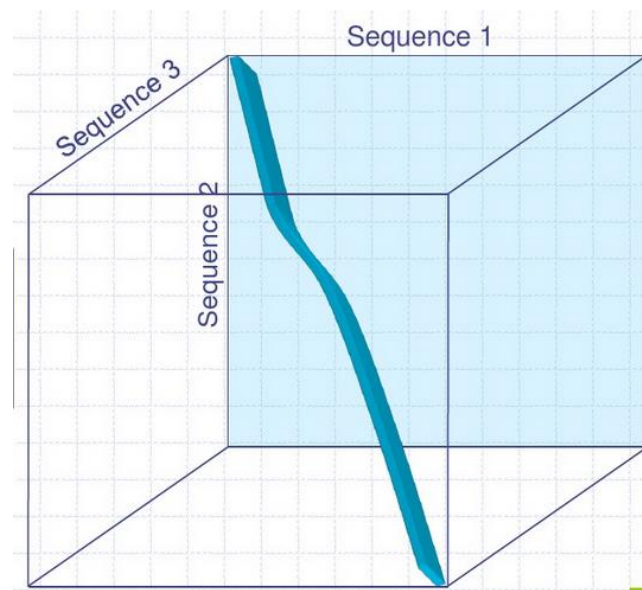
通过特定的算法（如**同步法**, **渐进法**等），由计算机程序自动搜索最佳的多序列比对状态。



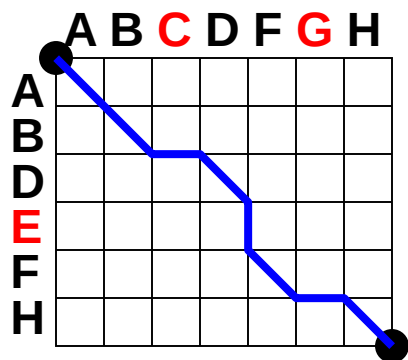
自动多序列比对的算法

1. 同步法

- 将序列两两比对时的**二维动态规划矩阵**扩展到**三维矩阵**。
- 即用矩阵的维数来反映比对的序列数目。
- 这种方法的计算量很大，对于计算机系统的资源要求比较高。

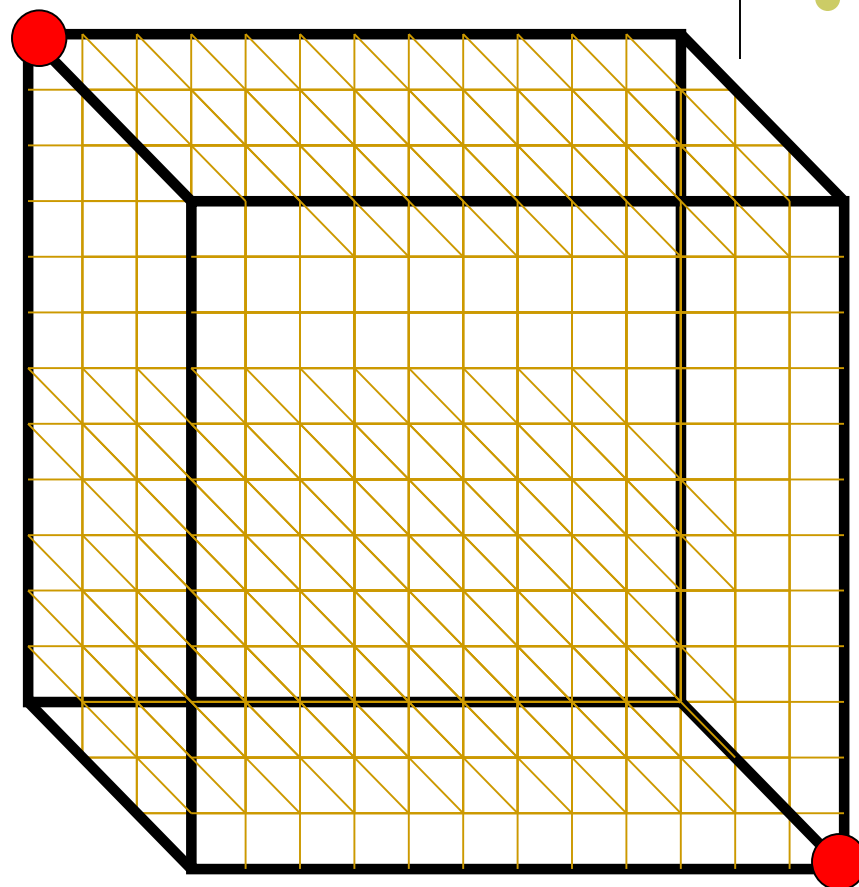


2-D vs 3-D 比对网格



2-D 矩阵

ABCD-FGH
AB-DEF-H



3-D 矩阵



回顾双序列比对算法



C	9																			
S	-1	4																		
T	-1	1	5																	
P	-3	-1	-1	7																
A	0	1	0	-1	4															
G	-3	0	-2	-2	0	6														
N	-3	1	0	-2	-2	0	6													
D	-3	0	-1	-1	-2	-1	1	6												
E	-4	0	-1	-1	-1	-2	0	2	5											
Q	-3	0	-1	-1	-1	-2	0	0	2	5										
H	-3	-1	-2	-2	-2	-2	1	-1	0	0	8									
R	-3	-1	-1	-2	-1	-2	0	-2	0	1	0	5								
K	-3	0	-1	-1	-1	-2	0	-1	1	1	-1	2	5							
M	-1	-1	-1	-2	-1	-3	-2	-3	-2	0	-2	-1	-1	5						
I	-1	-2	-1	-3	-1	-4	-3	-3	-3	-3	-3	-3	-3	1	4					
L	-1	-2	-1	-3	-1	-4	-3	-4	-3	-2	-3	-2	-2	2	2	4				
V	-1	-2	0	-2	0	-3	-3	-3	-2	-2	-3	-3	-2	1	3	1	4			
F	-2	-2	-2	-4	-2	-3	-3	-3	-3	-3	-1	-3	-3	0	0	0	-1	6		
Y	-2	-2	-2	-3	-2	-3	-2	-3	-2	-1	2	-2	-2	-1	-1	-1	-1	3	7	
W	-2	-3	-2	-4	-3	-2	-4	-4	-3	-2	-2	-3	-3	-1	-3	-2	-3	1	2	11
	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W

序列比对：

VDSCY

和

VESLCY

空位罚分d=

序列比对：
VDSCY
和
VESLCY

空位罚分 $d=-11$

BLOSUM62

构建配对得分矩阵



	V	D	S	C	Y
V	4	-3	-2	-1	-1
E	-2	2	0	-4	-2
S	-2	0	4	-1	-2
L	1	-4	-2	-1	-1
C	-1	-3	-1	9	-2
Y	-1	-3	-2	-2	7

构建最佳上游节点矩阵

	V	D	S	C	Y
V	$S_{i-1, j-1}$	$S_{i-1, j}$			
E	$S_{i, j-1}$	$S_{i, j}$			
S					
L					
C					
Y					

↓ D-
V-E

↘ VD
VE

→ VD
E-

要求解 S_{ij} 的分数，我们必须先知道

$S_{i-1, j-1}$, $S_{i-1, j}$, $S_{i, j-1}$ 的分数，这种方法叫做**递归算法**；

采用这种方法，可以把大的问题分割成小的问题逐一解决，即**动态规划算法**；需要存储如何得到 S_{ij} 分数的过程。



构建最佳上游节点矩阵

	V	D	S	C	Y
V	$S_{i-1, j-1}$	$S_{i-1, j}$			
E	$S_{i, j-1}$	$S_{i, j}$			
S					
L					
C					
Y					

Needleman-Wunsch算法;

$$S_{ij} = \max \begin{cases} S_{i-1, j-1} + \sigma(x_i, y_j) \\ S_{i-1, j} + d \text{ (从上到下)} \\ S_{i, j-1} + d \text{ (从左到右)} \end{cases}$$



构建最佳上游节点矩阵

	Gap	V	D	S	C	Y
Gap	0	-11				
V	-11	4				
E						
S						
L						
C						
Y						

Needleman-Wunsch算法;

$$S_{ij} = \max \begin{cases} S_{i-1,j-1} + \sigma(x_i, y_j) \\ S_{i-1,j} + d \text{ (从上到下)} \\ S_{i,j-1} + d \text{ (从左到右)} \end{cases}$$

配对得分矩阵



	V	D	S	C	Y
V	4	-3	-2	-1	-1
E	-2	2	0	-4	-2
S	-2	0	4	-1	-2
L	1	-4	-2	-1	-1
C	-1	-3	-1	9	-2
Y	-1	-3	-2	-2	7



构建最佳上游节点矩阵

	Gap	V	D	S	C	Y
Gap	0	-11	-3-22			
V	-11	4	-7			
E						
S						
L						
C						
Y						

$V \leftrightarrow D: -3$



最佳上游节点矩阵

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

Diagram illustrating the Best Upstream Node Matrix (最佳上游节点矩阵) with values and arrows indicating relationships between nodes.

Nodes: Gap, V, D, S, C, Y.

Values (Matrix):

- Gap: 0, -11, -22, -33, -44, -55
- V: -11, 4, -7, -18, -29, -40
- E: -22, -7, 6, -5, -16, -27
- S: -33, -18, -5, 10, -1, -12
- L: -44, -29, -16, -1, 9, -2
- C: -55, -40, -27, -12, 8, 7
- Y: -66, -51, -38, -23, -3, 15

Arrows and Green Numbers (Relationships):

- Gap to V: 4
- V to D: 2
- D to S: 4
- S to C: -1
- C to Y: -2
- Y to Gap: 9
- Y to C: 7



回溯

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

Diagram illustrating a backtracking search for a path through a grid of values. The grid is 7x7, with rows labeled Gap, V, E, S, L, C, Y and columns labeled Gap, V, D, S, C, Y. The values are as follows:

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

Red arrows indicate the path taken during the search, starting from 0 and ending at 15. Green numbers (4, 2, 4, -1, 9, 7) are placed near the arrows, likely representing a cost or weight. The final value 15 is highlighted in red.



回溯

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

Diagram illustrating a sequence of values and transitions (arrows) across a grid. The grid has columns labeled Gap, V, D, S, C, Y and rows labeled Gap, V, E, S, L, C, Y. Red arrows show a path from 0 to -11 to -22 to -33 to -44 to -55, and from -11 to -22 to -33 to -44 to -55. Green numbers (4, 2, 4, -1, 9, 7) are placed near the arrows. A blue arrow points from -3 to 15.



回溯

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

Diagram illustrating a sequence of values and transitions (arrows) between them, likely representing a dynamic programming or search process. The values are arranged in a grid with rows labeled Gap, V, E, S, L, C, Y and columns labeled Gap, V, D, S, C, Y. Red arrows indicate a path from 0 to -11 to -22 to -33 to -44 to -55, and from -11 to 4 to -7 to 6 to -5 to 10 to -1 to 9 to 8 to -3 to 15. Green numbers (4, 2, 4, -1, 9, 7) are placed near the arrows. A blue arrow points from -1 to 8. A red arrow points from 9 to 7.



回溯

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

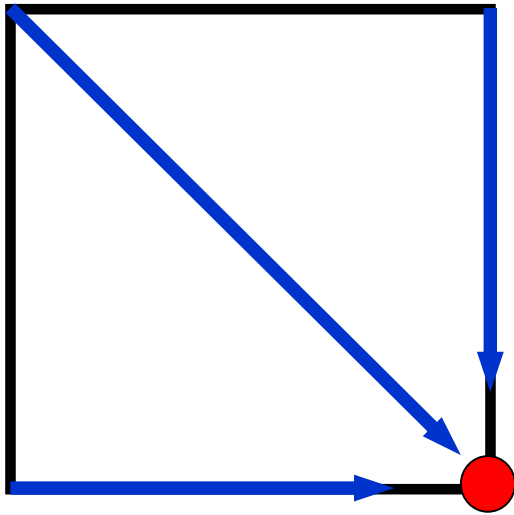
Diagram illustrating a sequence of values and transitions (arrows) across a grid. The grid has columns labeled Gap, V, D, S, C, Y and rows labeled Gap, V, E, S, L, C, Y. Red arrows indicate a path from (Gap, Gap) to (Y, Y) with values: 0 → -11 → -22 → -33 → -44 → -55 → -66. Blue arrows indicate a path from (Gap, Gap) to (Y, Y) with values: 0 → 4 → -7 → 6 → -5 → 10 → -1 → 9 → 8 → -3 → 15. Green numbers (4, 2, 4, -1, 9, 7) are placed near the blue arrows, and a green number (2) is placed near the red arrow from -11 to -7.



比对结果:

V D S - C Y
V E S L C Y

	Gap	V	D	S	C	Y
Gap	0	-11	-22	-33	-44	-55
V	-11	4	-7	-18	-29	-40
E	-22	-7	6	-5	-16	-27
S	-33	-18	-5	10	-1	-12
L	-44	-29	-16	-1	9	-2
C	-55	-40	-27	-12	8	7
Y	-66	-51	-38	-23	-3	15

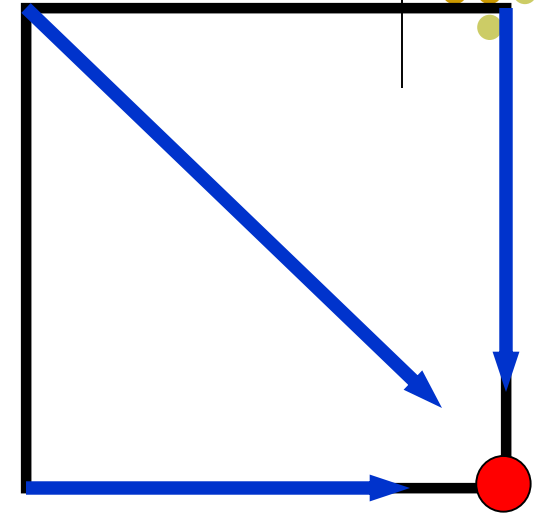
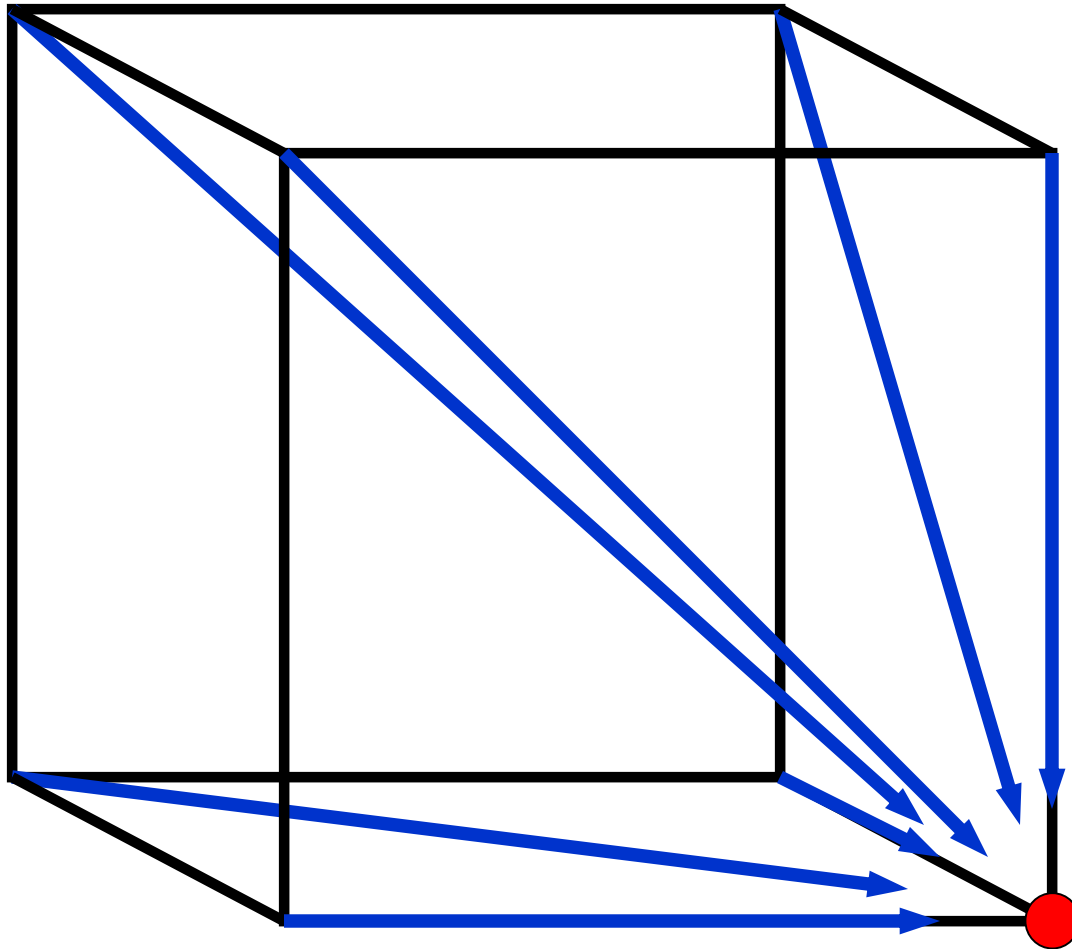


In **2-D**, 3 edges
in each unit
square

Needleman-Wunsch算法 ;

$$S_{ij} = \max \begin{cases} S_{i-1, j-1} + \sigma(x_i, y_j) \\ S_{i-1, j} + d \text{ (从上到下)} \\ S_{i, j-1} + d \text{ (从左到右)} \end{cases}$$

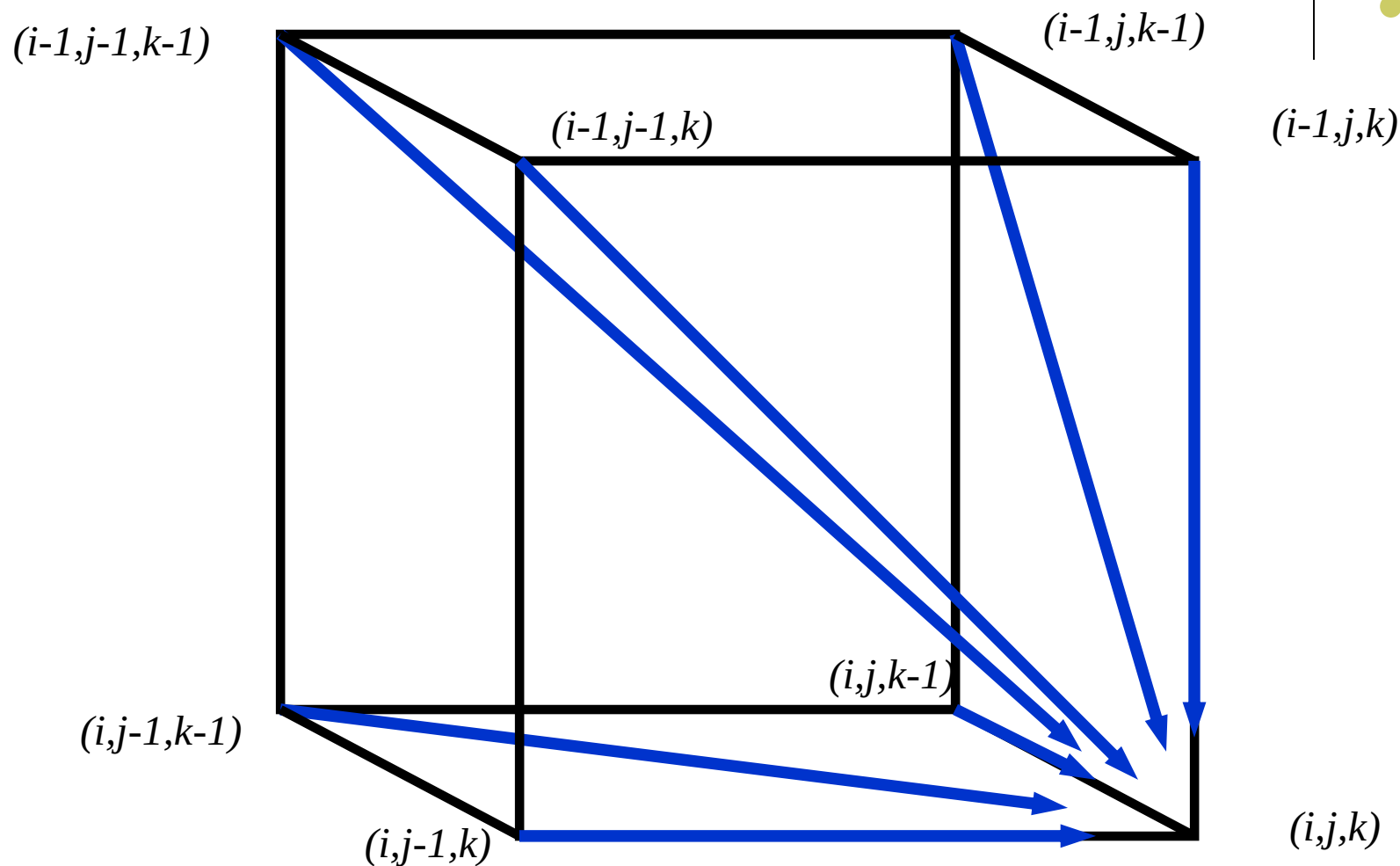
2-D cell versus 3-D Alignment Cell



In **2-D**, 3 edges
in each unit
square

In **3-D**, 7 edges
in each unit cube

3-D 比对单元的结构





Multiple Alignment: Dynamic Programming

- $s_{i,j,k} = \max \begin{cases} s_{i-1,j-1,k-1} + \delta(v_i, w_j, u_k) \\ s_{i-1,j-1,k} + \delta(v_i, w_j, _) \\ s_{i-1,j,k-1} + \delta(v_i, _, u_k) \\ s_{i,j-1,k-1} + \delta(_, w_j, u_k) \\ s_{i-1,j,k} + \delta(v_i, _, _) \\ s_{i,j-1,k} + \delta(_, w_j, _) \\ s_{i,j,k-1} + \delta(_, _, u_k) \end{cases}$

- $\delta(x, y, z)$ 是 3-D 打分矩阵中的元素。

如果比对k条蛋白质序列，记分矩阵是？迭代公式的方程个数是？



多序列比对运行时间

- 对于3条长 n 的序列，算法的时间复杂度是 $7n^3$;
 $O(n^3)$
- 对于 k 条序列，时间复杂度是 $(2^k-1)(n^k)$; $O(2^k n^k)$
- 结论: 动态规划法虽然可以直接从2维问题推广到高维问题，但是算法是**指数复杂度**的。
- **同步法 不适用于大规模多序列比对!**

自动多序列比对的算法



2. 步进法

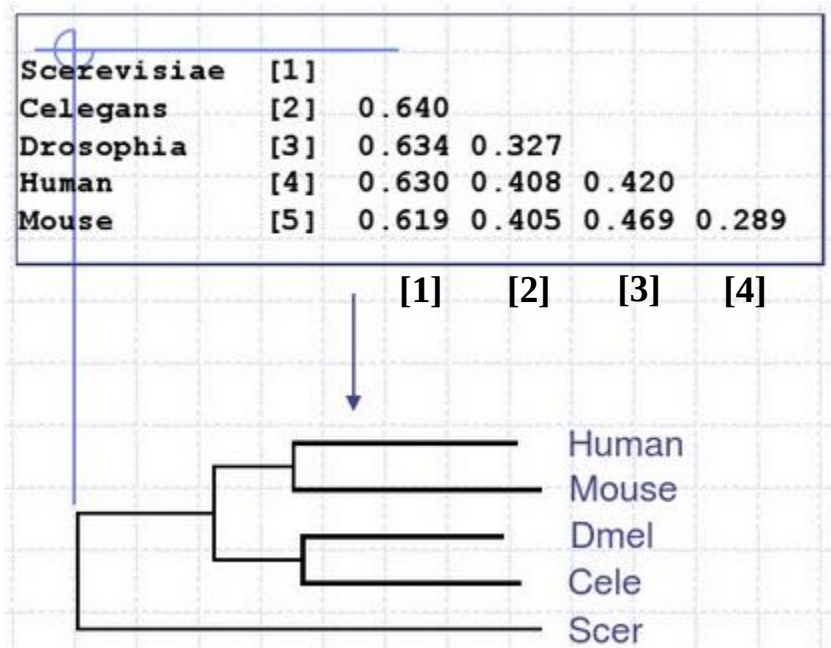
- 贪婪算法。
- 选择最相似的两条序列构建相似序列，则序列总数变成了 $k-1$ 。
- 重复这个过程，一直到最后所有序列完成比对。

<i>s1</i>	GTCTGA
<i>s2</i>	GATTCA
<i>s3</i>	GTCAGC
<i>s4</i>	GATATT
<i>s5</i>	GCTCCA

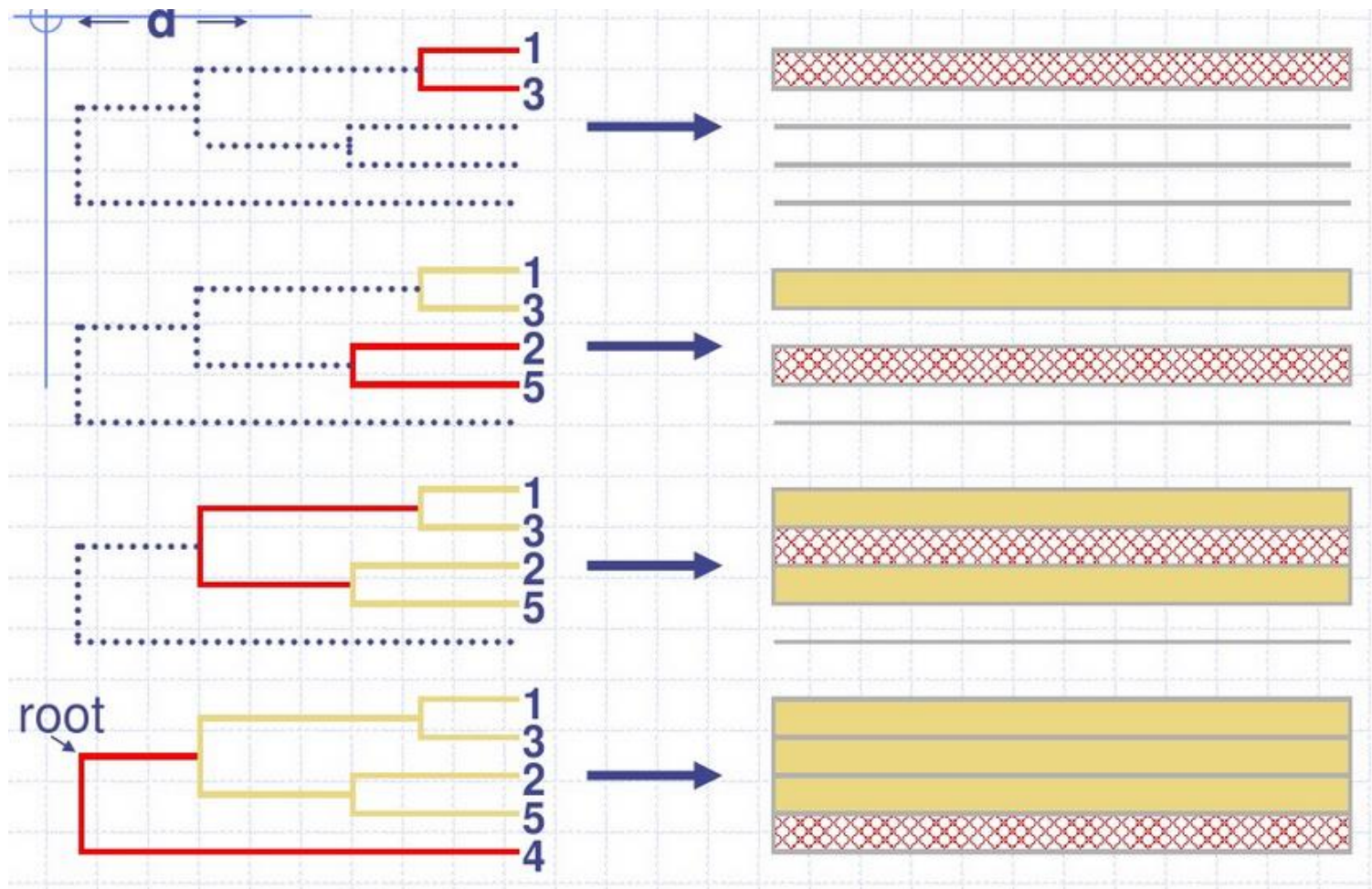
自动多序列比对的算法



s1 GTCTGA
s2 GATTCA
s3 GTCAGC
s4 GATATT
s5 GCTCCA



自动多序列比对的算法



如何合并最近序列？



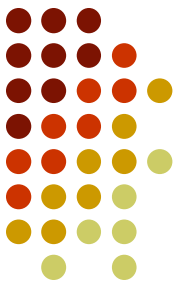
s1 GTCTGA

s2 GATTCA

s3 GTCAGC

s4 GATATT

s5 GCTCCA



S_1 和 S_3 最相似;

s_1 **GTC**T**G**A
 s_3 **GTC**A**G**C } **GTC** t/a **G** a/c

5条“序列”变成4条:

$S_{1,3}$	GTC	W	G	M
S_2	GAT	T	C	A
S_4	GAT	A	T	T
S_5	GCT	C	C	A

代码	
G	
A	
T	(U)
C	
R	(A or G)
Y	(C or T or U)
M	(A or C)
K	(G or T)
S	(C or G)
W	(A or T)
H	(A or C or T)
B	(C or G or T)
V	(A or C or G)
D	(A or G or T)
N	(A or C or G or T)

多序列比对工具



软件名称	特点简介	适用场景
Clustal Omega	快速、适用于大规模比对，支持命令行和Web版本	蛋白质/核酸序列初步比对
MAFFT	多种算法选择（FFT、L-INS-i等），速度快且准确性高	高质量比对、大数据集
MUSCLE	高准确率，适合中小规模数据	系统发育分析
T-Coffee	可以结合多个比对结果，提高一致性	比对结果可靠性提升
ProbCons	基于概率一致性的蛋白质比对方法，准确性较高	蛋白质序列比对（较慢）
Kalign	基于Wu-Manber算法，运行速度极快	快速分析（中低准确度）
DIALIGN-TX	无需全局同源，适合序列部分区域保守的情况	局部区域对齐
MSAProbs	基于概率模型 + 并行处理，性能优秀	蛋白质序列高精度对齐



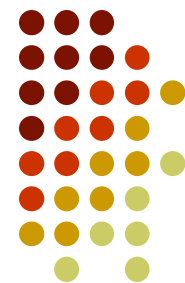
多序列比对工具

— clustal

Clustal是一个单机版的基于渐进比对的多序列比对工具，由Higgins D.G. 等开发。

有应用于多种操作系统平台的版本，包括linux版，DOS版的clustlw，windows版本的clustalx等。

Clustal简介

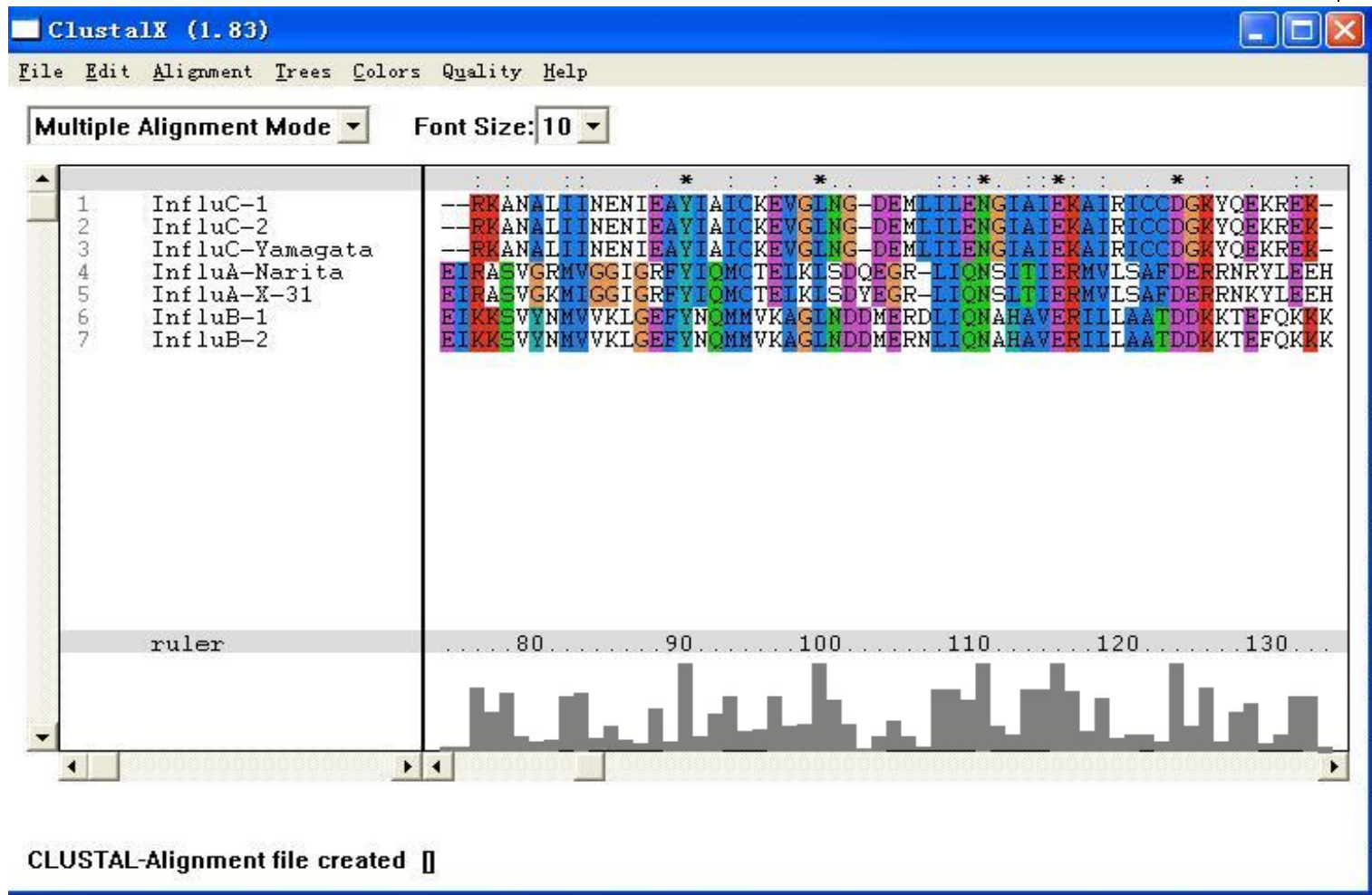


- 算法：渐进比对方法

1. 将多个序列两两比对构建**距离矩阵**，反应序列之间两两关系；
2. 根据距离矩阵生成**系统进化指导树**；
3. 从**最紧密**的两条序列开始，**逐步引入临近的序列**并不断重新构建比对，直到所有序列都被加入为止。



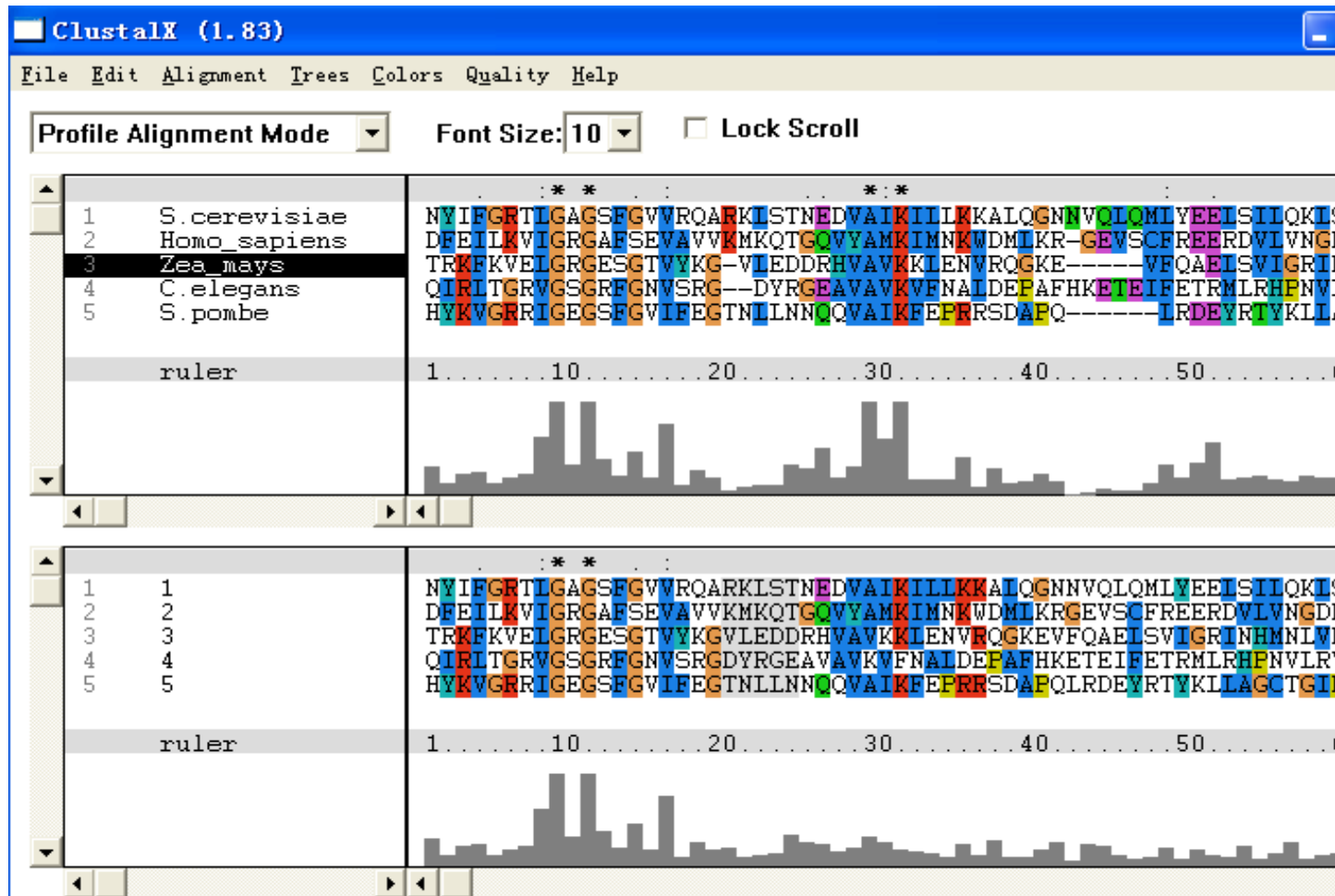
Clustalx的工作界面 (多序列比对模式)





Clustalx的工作界面

(剖面(profile)比对模式)



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Clustal的工作原理



Clustal输入多个序列



快速的序列两两比对，计算序列间的距离，获得一个距离矩阵。

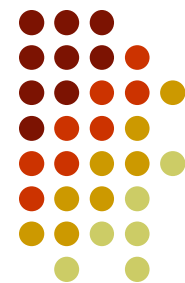


邻接法(**NJ**)构建一个树（引导树）



根据引导树，渐进比对多个序列。

Clustal的应用



1.输入输出格式。

输入序列的格式比较灵活，可以是前面介绍过的**FASTA**格式，还可以是PIR、SWISS-PROT、GDE、Clustal、GCG/MSF、RSF等格式。

输出格式也可以选择，有**ALN**、GCG、PHYLIP和NEXUS等，用户可以根据自己的需要选择合适的输出格式。

Clustal的应用



2.两种工作模式。

a.多序列比对模式。

b.剖面(profile)比对模式。

3.一个实际的例子。



多序列比对实例

输入文件的格式(fasta):

```
>KCC2_YEAST
NYIFGRTLGAAGSFGVVRQARKLSTN.....
>DMK_HUMAN
DFEILKVIGRGAFSEVAVVKMKQTGQVYAMKIMNK.....
>KPRO_MAIZE
TRKFKVELGRGESGTVYKGVLEDDRHVAVKKLEN.....
>DAF1_CAEEL
QIRLTGRVGSGRFGNVSRGDYRGEAVAVKVFNALD.....
>1CSN
HYKVGRRIGEGSFGVIFEGTNLLNN.....
```


第一步：输入序列文件。



ClustalX (1.83)

File Edit Alignment Trees Colors Quality Help

Font Size: 10

Load Sequences
Append Sequences
Save Sequences as...
Load Profile 1
Load Profile 2
Save Profile 1 as...
Save Profile 2 as...
Write Alignment as PostScript
Write Profile 1 as PostScript
Write Profile 2 as PostScript
Quit

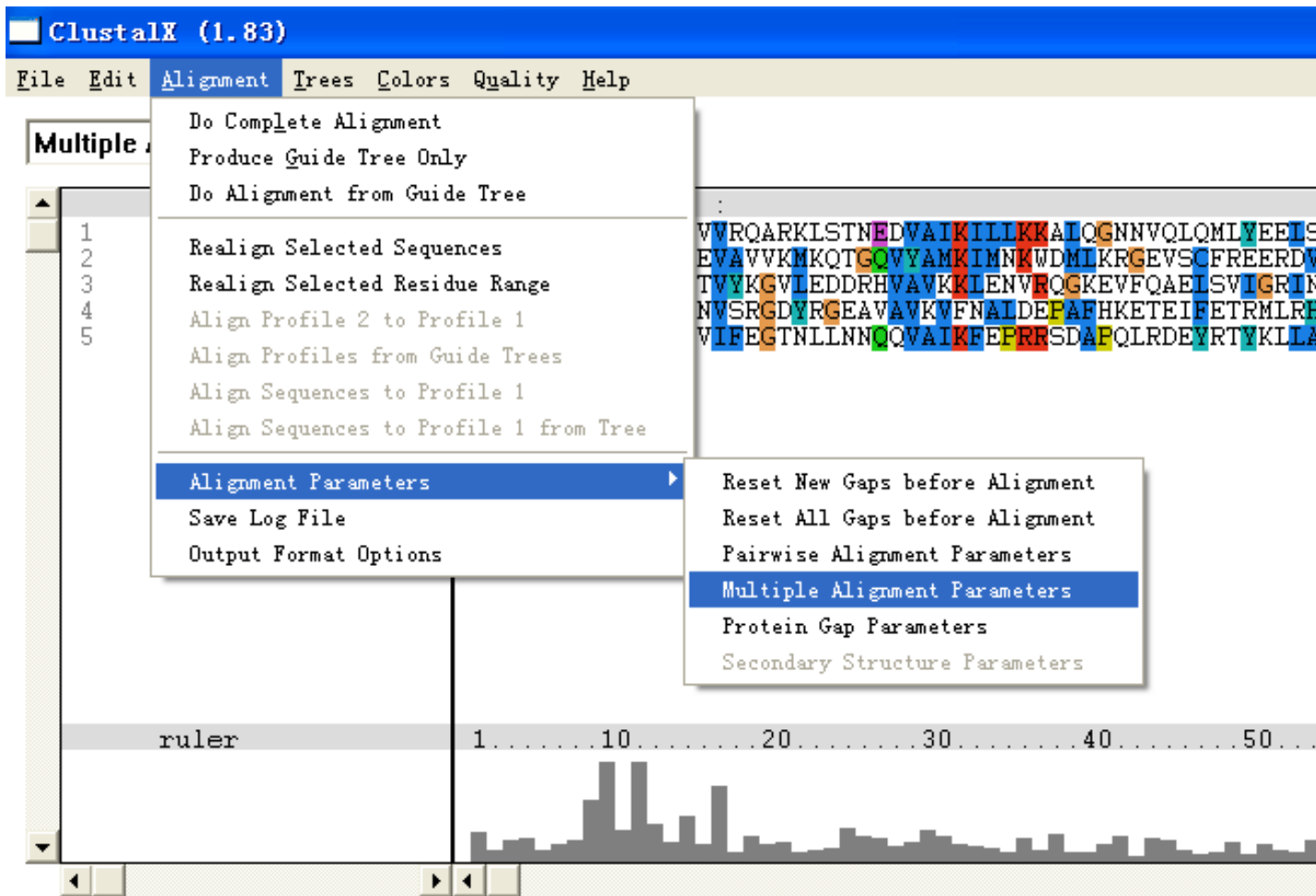
Sequence Alignment View:

Sequence 1: NVIFGRTLCAGSFGVVRQARKLSTNEDVAIKILLKKAIOGNNVQLQMLYEELSILO
Sequence 2: DFEILKVIKRGAFSEVAVVKMKQTGOVYAMKIMNKWDMIKRGEVSCFREERDVLVN
Sequence 3: TRKFKVELGRGESGTVYKGVLEDDRHVAVKKLENVROGKEVFQAEISVIGRINHMN
Sequence 4: QIRLTGRVGSGRFGNVSRGDYRGEAVAVKVFNALDEFAPHKETEIFETRMLRHFNV
Sequence 5: HYKVGRRIGEGSFGVIFEGTNLLNNQVAKFEPRRSDAPQLRDEVRTYKLLAGCT

ruler 1 10 20 30 40 50

File D:\1.seq loaded.

第二步：设定比对的一些参数。



File D:\1.seq loaded.



参数设定窗口。

Alignment Parameters

CLOSE

Multiple Parameters

Gap Opening [0-100] : 10.00 Gap Extention [0-100] : 0.20

Delay Divergent Sequences (%) : 30

DNA Transition Weight [0-1] : 0.50

Use Negative Matrix OFF ▼

Protein Weight Matrix

☐ BLOSUM series ☐ PAM series

☒ Gonnet series ☐ Identity matrix

☐ User defined

Load protein matrix:

DNA Weight Matrix

☒ IUB ☐ CLUSTALW(1.6)

☐ User defined

Load DNA:

第三步：开始序列比对。



ClustalX (1.83)

File Edit Alignment Trees Colors Quality Help

Multiple

- Do Complete Alignment
- Produce Guide Tree Only
- Do Alignment from Guide Tree
- Realign Selected Sequences
- Realign Selected Residue Range
- Align Profile 2 to Profile 1
- Align Profiles from Guide Trees
- Align Sequences to Profile 1
- Align Sequences to Profile 1 from Tree
- Alignment Parameters
- Save Log File
- Output Format Options

1
2
3
4
5

WVRQARKLSTNEDVAIKILLKKALQGNNVQLQMLVEELIS
EVAVVKMKQTGOVYAMKIMNKWDMLKRGVSCFREERDV
TVYKGVLEDDRHVAVKKIENVFQKKEVFQAEISVIGRIK
NVSRGDYRGEAVAVKVFNAIDEPAFHKETEIFETRMLRE
WIFEGTNLLNNQVAIKFEPRRSDAFQLRDEVRTYKLLA

ruler 1 10 20 30 40 50

File D:\1.seq loaded.

ClustalX (1.83)

File Edit Alignment Trees Colors Quality Help

Multiple Alignment Mode Font Size: 10

Complete Alignment

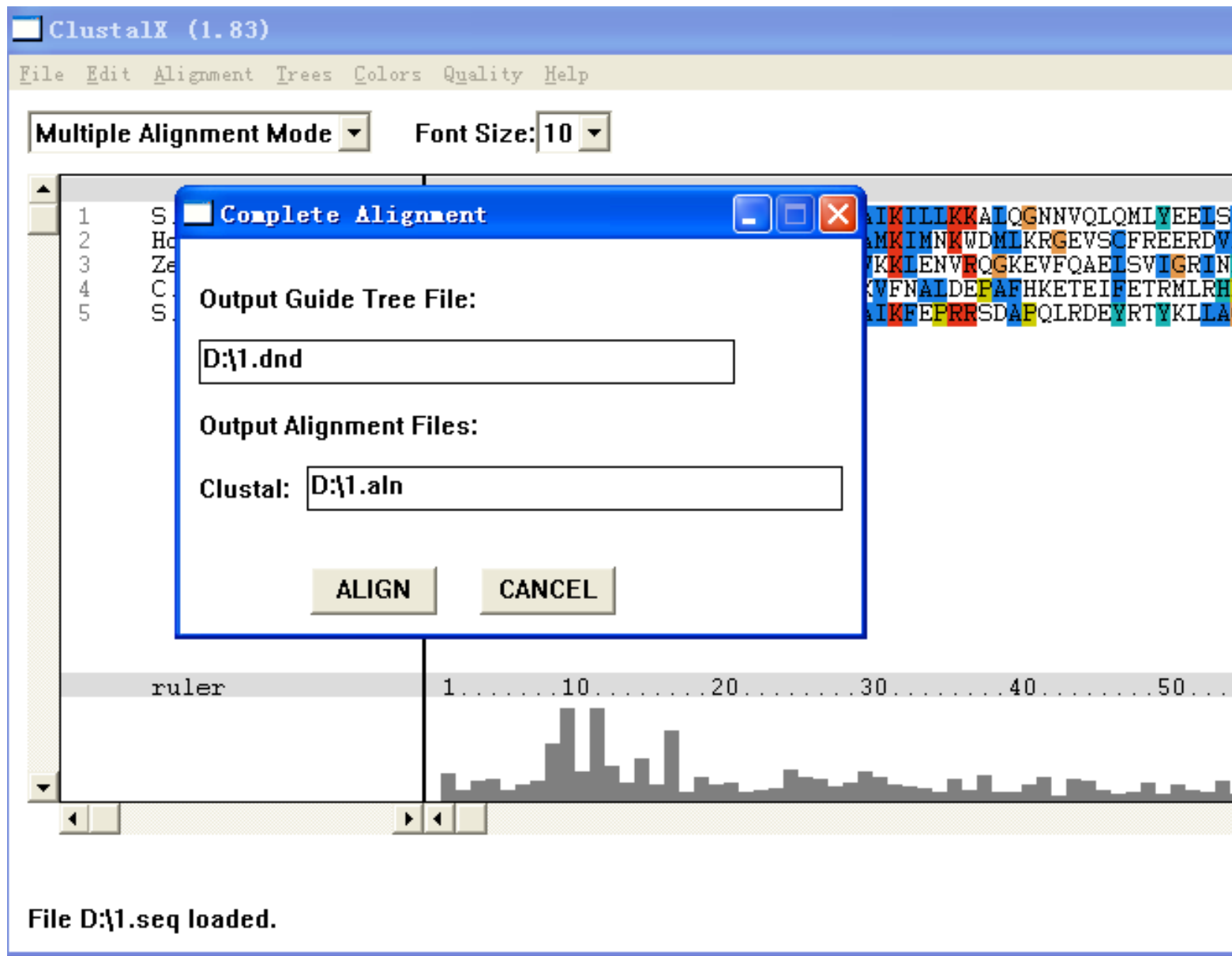
Output Guide Tree File:
D:\1.dnd

Output Alignment Files:
Clustal: D:\1.aln

ALIGN CANCEL

ruler 1 10 20 30 40 50 ...

File D:\1.seq loaded.



The image shows the ClustalX (1.83) software interface. At the top is a menu bar with 'File', 'Edit', 'Alignment', 'Trees', 'Colors', 'Quality', and 'Help'. Below the menu bar is a toolbar with 'Multiple Alignment Mode' and 'Font Size: 10'. The main window displays a sequence alignment with a ruler at the bottom. A 'Complete Alignment' dialog box is open in the center, containing fields for 'Output Guide Tree File' (D:\1.dnd) and 'Output Alignment Files' (Clustal: D:\1.aln), along with 'ALIGN' and 'CANCEL' buttons. The background shows a sequence alignment with a ruler and a histogram.

第四步：比对完成，选择保存结果文件的格式



ClustalX (1.83)

File Edit Alignment Trees Colors Quality Help

Font Size: 10

Load Sequences
Append Sequences
Save Sequences as...
Load Profile 1
Load Profile 2
Save Profile 1 as...
Save Profile 2 as...
Write Alignment as PostScript
Write Profile 1 as PostScript
Write Profile 2 as PostScript
Quit

Alignment view showing 5 sequences (NYIFGRTLGAGSFGVVRQARKLSTNEDVAIKILLKKALQGNVQLQMLYEE, DFEILKVIKRGAFSEVAVVMKQTGOVYAMKIMNKWDMIKR-GEVSCFREE, TRKFKVELGRGESGTVYKG-VLEDDRHVAVKKLENVRQKE----VFQAE, QIRITGRVGSGRFCNVSRG--DYRGEAVAVKVENALDEFAFHKETEIFETR, HYKVGRIRIGESFCVIFECTNLLNNQOVAIKFEFERSDAFQ-----LRDE) with a ruler and a histogram below.

CLUSTAL-Alignment file created []

 SAVE SEQUENCES



Format

- ☐ CLUSTAL ☐ NBRF/PIR ☐ GCG/MSF
☒ **PHYLIP** ☐ GDE ☐ NEXUS
☐ FASTA

GDE output case : **Lower** ▼

CLUSTALW sequence numbers : **OFF** ▼

Save range from : **0** to : **0** and include range numbers : **OFF** ▼

SAVE SEQUENCES AS:

D:\1.phy

OK

CANCEL



在线的**clustalw**分析

1. **EBI**提供的在线**clustalw**服务

<http://www.ebi.ac.uk/clustalw/>

EBI提供的 在线 Clustalw 服务

ClustalW - Microsoft Internet Explorer

文件(F) 编辑(E) 查看(V) 收藏(A) 工具(T) 帮助(H)

后退 搜索 收藏夹 媒体

地址 http://www.ebi.ac.uk/clustalw/ 转到

EBI Home About EBI Research Services **Toolbox** Databases Downloads Submission

SEQUENCE ANALYSIS

- Help Index
- General Help
- Formats
- Gaps
- Matrix
- References
- ClustalW Help

ClustalW Submission Form

Clustal W is a general purpose multiple sequence alignment program for DNA or proteins. It produces biologically meaningful multiple sequence alignments of divergent sequences. It calculates the best match for the selected sequences, and lines them up so that the identities, similarities and differences can be seen. Evolutionary relationships can be seen via viewing Cladograms or Phylograms.

[Download Software](#)

ALIGNMENT TITLE	ALIGNMENT	OUTPUT FORMAT	OUTPUT ORDER	COLOR ALIGNMENT	
-NONE-	full	aln w/numbers	aligned	no	
KTUP (WORD SIZE)	WINDOW LENGTH	SCORE TYPE	TOPDIAG	PAIRGAP	
def	def	percent	def	def	
MATRIX	GAP OPEN	END GAPS	GAP EXTENSION	GAP DISTANCES	CPU MODE
def	def	def	def	def	single

TREE GRAPH		PHYLOGENETIC TREE		
TYPE	DISTANCES	TREE TYPE	CORRECT DIST.	IGNORE GAPS
cladogram	hide	none	off	off

Enter or Paste a set of Sequences in any supported format: [Help](#)





更为详细的教程

可以在这里得到更多关于**clustal**的帮助:

<http://www-igbmc.u-strasbg.fr/BioInfo/ClustalX/Top.html>